Math 110 — Week 6

Invariant subspaces, eigenvalues and eigenvectors, PageRank (Notes Ch. 11, LADR 5A, HW6)

§1 Facts used from Weeks 1–5

(W1) In a space of dimension n, any $n+1$ vectors are dependent, and an independent list has at most n vectors. If $U \cap W = \{0\}$ and $\dim U + \dim W = \dim V < \infty$, then $V = U \oplus W$ (Week 2).

(W2) Dependence lemma: in a dependent list $v_1, \dots, v_m$ some v_k lies in $\operatorname{span}(v_1, \dots, v_{k-1})$ (Week 2).

(W3) For an operator T on a finite-dimensional space: injective $\iff$ surjective $\iff$ invertible. T is injective $\iff \operatorname{null} T = \{0\}$ (Weeks 3, 4).

(W4) $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ is invertible $\iff \det A = ad - bc \neq 0$; when $\det A = 0$ the rows of A are dependent (Week 4).

(W5) $\mathcal{M}(T)\mathcal{M}(v) = \mathcal{M}(Tv)$: applying T multiplies the coordinate column by the matrix (Week 3).

(W6) A stochastic matrix P has entries ≥ 0 and each column sums to 1. If p is a probability vector, so is Pp (Week 3).

(W7) If $T(U_1) \subseteq U_2$, then $v + U_1 \mapsto Tv + U_2$ is a well-defined linear map $V/U_1 \to W/U_2$ (Week 5).

(W8) Dual map: $(T'\varphi)(v) = \varphi(Tv)$; $(S+T)' = S' + T'$, $(\lambda T)' = \lambda T'$, $I' = I$; $U^0 = \{\varphi \in V' : \varphi|_U = 0\}$; for finite-dimensional spaces, T injective $\iff$ T' surjective. A nonzero functional on V has a null space of dimension $\dim V - 1$ (Week 5).

(W9) $\mathbb{F}^\infty$ is the space of sequences $(a_1, a_2, \dots)$; the right shift $R(a_1, a_2, \dots) = (0, a_1, a_2, \dots)$ is injective but not surjective (Weeks 1, 4).

(W10) For complex numbers: $\overline{z+w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\bar{w}$, $z\bar{z} = a^2 + b^2$ for $z = a + bi$ (Week 1).

§2 This week

§2.1 Invariant subspaces, eigenvectors, eigenspaces (Ch. 11.1)

The question. $N = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ maps $\mathbb{R}^2$ to itself. Which lines through 0 does N map into themselves? Can $\mathbb{R}^2$ be split as a direct sum of two such lines?

Definition (Def. 30 / LADR 5.2)**.** Let $T \in \mathcal{L}(V)$. A subspace $U \subseteq V$ is **invariant under** T if $Tu \in U$ for every $u \in U$.

- To check invariance it is enough to check $Tu_i \in U$ for a spanning list $u_1, \dots, u_k$ of U: then $T(\sum c_i u_i) = \sum c_i T u_i$ is a combination of vectors in U.
- If U is invariant, T gives two smaller operators: the **restriction** $T|_U : U \to U$, and the **quotient operator** $T/U : V/U \to V/U$, $(T/U)(v+U) = Tv + U$ (well defined by (W7) with $U_1 = U_2 = U$).

Example 48: subspaces that are always invariant. $\{0\}$, V, $\operatorname{null} T$ and $\operatorname{range} T$ are invariant under T. If $u \in \operatorname{null} T$, then $Tu = 0 \in \operatorname{null} T$. If $u \in \operatorname{range} T$, then $Tu \in \operatorname{range} T$ because Tu is an output of T.

Definition (Def. 31, 32 / LADR 5.5, 5.8, 5.52)**.** A *nonzero* vector v is an **eigenvector** of T if $Tv = \lambda v$ for some $\lambda \in \mathbb{F}$. The scalar λ is the **eigenvalue** of v. The **eigenspace** of λ is

$$E(\lambda, T) = \operatorname{null}(T - \lambda I) = \{v : Tv = \lambda v\}.$$

- **An eigenvector is the same as a one-dimensional invariant subspace.** If $\operatorname{span}(v)$ is invariant, then $Tv \in \operatorname{span}(v)$, so $Tv = \lambda v$. Conversely, if $Tv = \lambda v$, then $T(cv) = c\lambda v \in \operatorname{span}(v)$, so every vector cv of $\operatorname{span}(v)$ is mapped into $\operatorname{span}(v)$.
- λ is determined by v: if $\lambda v = \mu v$ with $v \neq 0$, then $(\lambda - \mu)v = 0$, so $\lambda = \mu$.
- $E(\lambda, T)$ consists of 0 and all eigenvectors with eigenvalue λ. It is a subspace (a null space), and it is invariant: if $Tv = \lambda v$, then $T(Tv) = T(\lambda v) = \lambda(Tv)$.
- λ is an eigenvalue of T exactly when $E(\lambda, T) \neq \{0\}$.

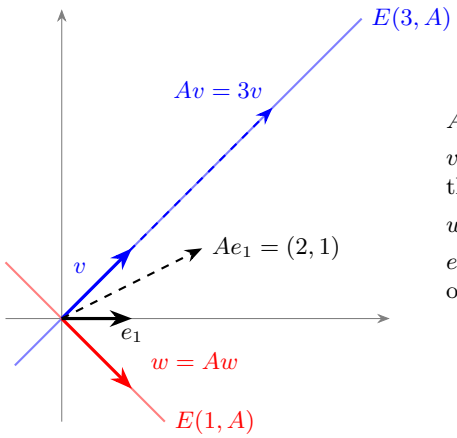
Proposition (LADR 5.7: four ways to say "λ is an eigenvalue")
Let V be finite-dimensional. The following are equivalent: (a) λ is an eigenvalue of T; (b) $T - \lambda I$ is not injective; (c) $T - \lambda I$ is not surjective; (d) $T - \lambda I$ is not invertible.

Proof. 1. **(a) $\iff$ (b).** $Tv = \lambda v$ is the same equation as $(T - \lambda I)v = 0$. So a nonzero v with $Tv = \lambda v$ exists exactly when $\operatorname{null}(T - \lambda I) \neq \{0\}$, that is, when $T - \lambda I$ is not injective (W3).

2. **(b) $\iff$ (c) $\iff$ (d).** $T - \lambda I$ is an operator on the finite-dimensional V, so it is injective $\iff$ surjective $\iff$ invertible (W3); negate each.

□

In particular, for V finite-dimensional: 0 **is an eigenvalue of** T $\iff$ T **is not invertible**, and $E(0, T) = \operatorname{null} T$. (The right shift R on $\mathbb{F}^\infty$ is not invertible and has no eigenvalue 0, since $Rv = 0$ only for $v = 0$.)



$A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$.
$v = (1, 1)$: $Av = (3, 3) = 3v$. The line through v is mapped into itself.
$w = (1, -1)$: $Aw = (1, -1) = w$.
$e_1 = (1, 0)$: $Ae_1 = (2, 1)$ is not a multiple of e_1. The line through e_1 is not invariant.

Eigenvectors are the vectors that A maps to multiples of themselves. Solid arrows are vectors, dashed arrows their images, all drawn at the same scale.

Example 49: all invariant subspaces of three operators on $\mathbb{R}^2$. In $\mathbb{R}^2$ the subspaces are $\{0\}$, lines, and $\mathbb{R}^2$. An invariant line is $\operatorname{span}(v)$ with v an eigenvector. So finding invariant subspaces means finding eigenvectors.

1. $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right]$: $(x, y) \mapsto (x, 2y) = \lambda(x, y)$ means $x = \lambda x$ and $2y = \lambda y$; for $(x, y) \neq 0$ this needs $y = 0$ ($\lambda = 1$) or $x = 0$ ($\lambda = 2$). Invariant: $\{0\}$, $\operatorname{span}(e_1)$, $\operatorname{span}(e_2)$, $\mathbb{R}^2$.
2. $\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ (rotation by 90°): $(-y, x) = \lambda(x, y)$ gives $-y = \lambda x$ and $x = \lambda y$, so $x = -\lambda^2 x$ and $y = -\lambda^2 y$; since $1 + \lambda^2 > 0$ for real λ, $x = y = 0$. Invariant: only $\{0\}$ and $\mathbb{R}^2$.

3. $N = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$: $N(x, y) = (y, 0) = \lambda(x, y)$ means $y = \lambda x$ and $0 = \lambda y$. If $\lambda \neq 0$, the second gives $y = 0$, and then the first gives $x = 0$: not an eigenvector. So $\lambda = 0$, and then $y = \lambda x = 0$. Invariant: $\{0\}$, $\operatorname{span}(e_1)$, $\mathbb{R}^2$. The line $\operatorname{span}(e_2)$ is not invariant because $Ne_2 = e_1$.

For N, $\operatorname{null} N = \operatorname{range} N = \operatorname{span}(e_1)$ (Week 3), and the chain $\{0\} \subset \operatorname{span}(e_1) \subset \mathbb{R}^2$ consists of invariant subspaces. A basis whose initial spans $\operatorname{span}(v_1) \subset \operatorname{span}(v_1, v_2) \subset \dots$ are all invariant is studied next week.

$$e_2 \xrightarrow{N} e_1 \xrightarrow{N} 0$$

N sends e_2 to e_1 and e_1 to 0. So $\operatorname{span}(e_1)$ is invariant and $\operatorname{span}(e_2)$ is not.

Stationary distributions are eigenvectors. For the stochastic matrix of Week 3,

$$P = \begin{bmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{bmatrix},$$

the probability vector $\pi = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ satisfies $P\pi = \pi$: it is an eigenvector with eigenvalue 1. A probability vector with $P\pi = \pi$ is called a **stationary distribution**. Then $P^k\pi = \pi$ for all k: apply P repeatedly, $P^{k+1}\pi = P(P^k\pi) = P\pi = \pi$. Four separate questions: does one exist, is it unique, does $P^k p_0$ converge to it, and how fast.

The question answered. N maps exactly one line into itself, $\operatorname{span}(e_1)$ (Ex. 49). So $\mathbb{R}^2$ is not a direct sum of two invariant lines.

Shortcuts

- **Invariant subspace check: apply T to a spanning list.** Reason: the first bullet after Def. 30. Instance: $\operatorname{span}(e_1)$ under N: $Ne_1 = 0 \in \operatorname{span}(e_1)$.
- **0 is an eigenvalue $\iff$ T is not invertible.** Reason: LADR 5.7 with $\lambda = 0$. Condition: V finite-dimensional. Instance: N is not invertible, and $Ne_1 = 0$.
- **Commuting operators.** If $ST = TS$, then $E(\lambda, T)$, $\operatorname{null} T$ and $\operatorname{range} T$ are invariant under S. Reason: if $Tv = \lambda v$, then $T(Sv) = S(Tv) = \lambda Sv$; if $Tv = 0$, then $T(Sv) = S(0) = 0$; and $S(Tu) = T(Su) \in \operatorname{range} T$. Instance: Midterm 2 extra practice (c), R6.
- **An equation for T restricts the eigenvalues.** If $Tv = \lambda v$, then $T^2 v = \lambda^2 v$. Reason: $T(Tv) = T(\lambda v) = \lambda Tv$. Instance: $T^2 = I$ forces $\lambda^2 = 1$ (Ex. 52 below).

Traps

- An eigenvector is nonzero by definition; an eigenvalue may be 0: N has the eigenvalue 0 with eigenvector e_1.
- The sum of two eigenvectors with different eigenvalues is not an eigenvector. For $\operatorname{diag}(1, 2)$, $e_1 + e_2 \mapsto (1, 2)$, which is not a multiple of $(1, 1)$. (Within one eigenspace, nonzero sums are eigenvectors.)
- An invariant subspace need not have an invariant complement: for N, the only invariant line is $\operatorname{span}(e_1)$.
- "0 is an eigenvalue $\iff$ T is not invertible" needs finite dimension: the right shift on $\mathbb{F}^\infty$ is not invertible, but 0 is not an eigenvalue.

§2.2 Complex numbers and the Fundamental Theorem of Algebra (Ch. 11.2)

The question. Does every operator have an eigenvector? Rotation of $\mathbb{R}^2$ by 90° has none (Ex. 49). Does the polynomial $z^2 + 1$ have a zero once we allow complex z, and does every nonconstant polynomial?

Facts about $\mathbb{C}$. For $z = a + bi$ let $|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}}$, the distance from z to 0.

- $|zw| = |z||w|$, since $|zw|^2 = zw\,\overline{zw} = z\bar{z}\,w\bar{w} = |z|^2|w|^2$ (W10).

- $|z+w| \le |z| + |w|$ (**triangle inequality**). For $u = a + bi$, $\operatorname{Re} u = a \le |u|$. So, using $|z\bar{w}| = |z||w|$,

$$|z+w|^2 = (z+w)(\bar{z}+\bar{w}) = |z|^2 + 2\operatorname{Re}(z\bar{w}) + |w|^2 \le |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2.$$

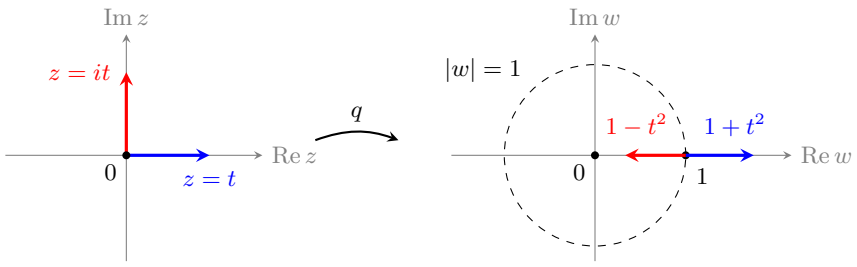
- **Polar form.** For $z \neq 0$, $z/|z|$ has absolute value 1, so it is a point $(\cos\theta, \sin\theta)$ of the unit circle: $z = r(\cos\theta + i\sin\theta)$ with $r = |z|$.
- **De Moivre.** By the angle-addition formulas,

$$(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta) = \cos(\alpha+\beta) + i\sin(\alpha+\beta).$$

  By induction on k, $z^k = r^k(\cos k\theta + i\sin k\theta)$. So every nonzero z has a k-th root:

$$\left(r^{1/k}\left(\cos\tfrac{\theta}{k} + i\sin\tfrac{\theta}{k}\right)\right)^k = r(\cos\theta + i\sin\theta) = z.$$

The key example: near $z = 0$, the polynomial $q(z) = 1 + z^2$ has $|q(0)| = 1$, and some directions make $|q|$ smaller.



inputs z near 0 values $q(z)$ near $q(0) = 1$

For $q(z) = 1 + z^2$: inputs along the real axis move q away from 0, outside the circle $|w| = 1$. Inputs along the imaginary axis have $z^2 = -t^2$, which points opposite to 1, so they move q toward 0, inside the circle. So $|q|$ has no minimum at 0.

Theorem (Prop. 52 / LADR 4.13: Fundamental Theorem of Algebra)
Every nonconstant polynomial with complex coefficients has a zero in $\mathbb{C}$.

The idea

Look at the size $|p(z)|$. Far from 0 it is large, so it has a smallest value somewhere, at a point ζ. We want to show that smallest value is 0. Suppose it is not. Then we find a nearby point where $|p|$ is even smaller, a contradiction.

Why can we always go lower from a nonzero value? Look at $q(z) = 1 + z^2$ at $z = 0$ (the figure above), where $|q| = 1$. Moving along the real axis, $q(t) = 1 + t^2$ gets bigger. Moving along the imaginary axis, $q(it) = 1 - t^2$ gets smaller. The term z^2 can be made to point in any direction by choosing the direction of z; choose it so that z^2 points opposite to the constant 1.

In general, after moving ζ to the origin and dividing by $p(\zeta)$, the polynomial is $q(z) = 1 + a_k z^k +$ (higher powers) with $a_k \neq 0$. Choose the direction β of z so that $a_k\beta^k = -1$ (possible because k-th roots exist in $\mathbb{C}$). Along $z = t\beta$ the polynomial is $1 - t^k$ plus terms with t^{k+1} and higher powers. For small t the term $-t^k$ is larger in absolute value than all the later terms together, so $|q(t\beta)| < 1$.

Proof. Let $p(z) = c_0 + c_1 z + \cdots + c_m z^m$ with $m \ge 1$, $c_m \neq 0$.

1. $|p(z)|$ **is large far away.** By the triangle inequality,

$$|p(z)| \ge |c_m||z|^m - \sum_{j<m} |c_j||z|^j,$$

   and the right side tends to ∞ as $|z| \to \infty$. So there is a radius R with $|p(z)| > |p(0)|$ whenever $|z| > R$.

2. **A global minimum exists.** On the closed disk $|z| \le R$ the continuous real function $|p|$ attains a minimum (extreme value theorem, cited from analysis), at some ζ. That minimum is $\le |p(0)|$, and outside the disk $|p|$ is larger than $|p(0)|$. So $|p(\zeta)| \le |p(z)|$ for all $z \in \mathbb{C}$.

3. **Suppose** $p(\zeta) \neq 0$. Let $q(z) = p(\zeta + z)/p(\zeta)$. Then $q(0) = 1$ and $|q(z)| \geq 1$ for all z.
4. q is a nonconstant polynomial, so $q(z) = 1 + a_k z^k + a_{k+1} z^{k+1} + \cdots + a_m z^m$, where $a_k z^k$ is the first nonzero term after the constant.
5. Choose $\beta \in \mathbb{C}$ with $\beta^k = -1/a_k$ (a k-th root). For real $t > 0$,
$$q(t\beta) = 1 - t^k + a_{k+1}\beta^{k+1} t^{k+1} + \cdots + a_m \beta^m t^m.$$
6. For $0 < t < 1$ the tail has absolute value at most Ct^{k+1}, where $C = \sum_{j>k} |a_j \beta^j|$. So
$$|q(t\beta)| \leq (1 - t^k) + Ct^{k+1} = 1 - t^k(1 - Ct).$$
7. For t small enough that $Ct < 1$, this is < 1. That contradicts $|q| \geq 1$. Hence $p(\zeta) = 0$.

□

Proposition (Factor theorem and factorisation)

If p is a polynomial over any field $\mathbb{F}$ and $p(\lambda) = 0$, then $p(z) = (z - \lambda)q(z)$ with $\deg q = \deg p - 1$. Over $\mathbb{C}$, every polynomial of degree $m \geq 1$ is $p(z) = c(z - \lambda_1) \cdots (z - \lambda_m)$, with the roots listed with multiplicity.

Proof. 1. Write $p(z) = \sum_{k=0}^{m} a_k z^k$. Then $p(z) - p(\lambda) = \sum_k a_k (z^k - \lambda^k)$, and
$$z^k - \lambda^k = (z - \lambda)(z^{k-1} + z^{k-2}\lambda + \cdots + \lambda^{k-1}).$$
2. So $p(z) = p(\lambda) + (z - \lambda)q(z)$ with q of degree $m - 1$ and leading coefficient a_m. If $p(\lambda) = 0$, this is the factorisation.
3. Over $\mathbb{C}$: p has a zero λ_1 (Prop. 52), so $p = (z - \lambda_1)q_1$ with $\deg q_1 = m - 1$. If $m - 1 \geq 1$, apply the same to q_1. After m steps the quotient is a constant c.

□

$\mathbb{C}$ is called **algebraically closed**. $\mathbb{R}$ is not: $x^2 + 1$ has no real zero, since $x^2 + 1 \geq 1$.

The question answered. Over $\mathbb{C}$, $z^2 + 1 = (z - i)(z + i)$ has the zeros $\pm i$, and every nonconstant complex polynomial has a zero (Prop. 52). Section 2.3 turns this into eigenvectors.

Shortcuts

- **k-th roots by polar form.** Write $z = r(\cos\theta + i \sin\theta)$ and take $r^{1/k}$ times the point at angle θ/k. Reason: De Moivre. Instance: i is the point at angle $\pi/2$, so the point at angle $\pi/4$ is a square root of i:
$$\cos\tfrac{\pi}{4} + i\sin\tfrac{\pi}{4} = \frac{1+i}{\sqrt{2}}, \qquad \left(\frac{1+i}{\sqrt{2}}\right)^2 = \frac{1 + 2i + i^2}{2} = i.$$
- **Nonreal roots of a real polynomial come in conjugate pairs.** If p has real coefficients, then $p(\bar{z}) = \overline{p(z)}$, so $p(\lambda) = 0 \Rightarrow p(\bar{\lambda}) = 0$. Reason: (W10). Instance: $t^2 + 1$ has i and $-i$.

Traps

- The Fundamental Theorem of Algebra is about polynomials with *complex* coefficients and positive degree: every such polynomial has a zero in $\mathbb{C}$. Over $\mathbb{R}$ it fails: $x^2 + 1$ has real coefficients and no real zero. (On Midterm 2 Q2(a) it was stated with "real coefficients", which is false.)
- A constant polynomial need not have a zero: $p(z) = 1$. "Nonconstant" is part of the statement.

§2.3 Existence and independence of eigenvectors (Ch. 11.2)

The question. Rotation by 90° on $\mathbb{C}^2$ sends $e_1 \mapsto e_2 \mapsto -e_1$, so $T^2e_1 + e_1 = 0$. From this one relation, can you produce a nonzero w and a $\lambda \in \mathbb{C}$ with $Tw = \lambda w$?

Definition (LADR 5.16: a polynomial applied to an operator)**.** For $p(z) = a_0 + a_1 z + \cdots + a_m z^m$ and $T \in \mathcal{L}(V)$ let $p(T) = a_0 I + a_1 T + \cdots + a_m T^m$, where T^k is T applied k times and $T^0 = I$.

Proposition (LADR 5.17, 5.18: polynomials in T)
(i) $(pq)(T) = p(T)q(T) = q(T)p(T)$. (ii) $\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are invariant under T. (iii) If $Tv = \lambda v$, then $p(T)v = p(\lambda)v$.

Proof. 1. **(i).** $T^iT^j = T^{i+j}$, and composition distributes over sums of operators. So multiplying out $p(T)q(T)$ term by term gives the same coefficients as multiplying out $p(z)q(z)$, which is $(pq)(T)$. Since $pq = qp$, also $q(T)p(T) = (pq)(T)$.

2. **(ii).** $T = z(T)$ for the polynomial z, so by (i) $Tp(T) = p(T)T$. If $p(T)v = 0$, then $p(T)(Tv) = T\,p(T)v = 0$. If $w = p(T)u$, then $Tw = p(T)(Tu) \in \operatorname{range} p(T)$.

3. **(iii).** $T^kv = \lambda^k v$ by induction: $T^0v = v$, and $T^{k+1}v = T(\lambda^k v) = \lambda^k Tv = \lambda^{k+1}v$. Multiply by a_k and add: $p(T)v = \sum a_k \lambda^k v = p(\lambda)v$.

□

Theorem (Prop. 53 / LADR 5.19: operators on complex spaces have eigenvectors)
Every operator on a nonzero finite-dimensional complex vector space has an eigenvector.

The idea

We need a polynomial so that the Fundamental Theorem can be used, and finite dimension produces one. Take any $v \neq 0$ and keep applying T: the vectors $v, Tv, T^2v, \ldots, T^nv$ are $n+1$ vectors in a space of dimension n, so they satisfy a relation $a_0v + a_1Tv + \cdots + a_mT^mv = 0$, that is, $p(T)v = 0$ for $p(z) = a_0 + a_1z + \cdots + a_mz^m$.

$p(T)v = 0$ does *not* say that v is an eigenvector. Instead factor $p(z) = (z - \lambda)q(z)$. By (i), $(T - \lambda I)\big(q(T)v\big) = p(T)v = 0$. So $w = q(T)v$ satisfies $Tw = \lambda w$. It is an eigenvector provided $w \neq 0$. To guarantee that, take the relation of *smallest* degree at the start: $q \neq 0$ (its leading coefficient is a_m) and q has smaller degree than p, so $q(T)v = 0$ would be a shorter relation.

Example (the question): T = rotation by 90° on $\mathbb{C}^2$, $v = e_1$. $Te_1 = e_2$, $T^2e_1 = -e_1$, so $T^2v + v = 0$: $p(z) = z^2 + 1 = (z - i)(z + i)$. Take $\lambda = i$, $q(z) = z + i$: $w = q(T)e_1 = Te_1 + ie_1 = (i, 1)$, and $Tw = (-1, i) = i \cdot (i, 1)$.

Proof. Let $n = \dim V$ and choose $v \neq 0$.

1. $v, Tv, \ldots, T^nv$ are $n+1$ vectors, so they are dependent (W1). Among all nontrivial relations

$$a_0v + a_1Tv + \cdots + a_mT^mv = 0, \qquad a_m \neq 0,$$

   choose one with m as small as possible. Then $m \geq 1$: if $m = 0$ the relation would be $a_0v = 0$ with $a_0 \neq 0$, so $v = 0$, which we excluded. So $p(z) = a_0 + \cdots + a_mz^m$ is nonconstant and $p(T)v = 0$.

2. By Prop. 52, p has a zero $\lambda \in \mathbb{C}$, so $p(z) = (z - \lambda)q(z)$ with $q(z) = b_0 + \cdots + b_{m-1}z^{m-1}$, $b_{m-1} = a_m \neq 0$ (factor theorem).

3. Let $w = q(T)v$, that is,

$$w = b_0v + b_1Tv + \cdots + b_{m-1}T^{m-1}v.$$

   Then $w \neq 0$: otherwise this would be a nontrivial relation of degree $m - 1 < m$.

4. By (i), $(T - \lambda I)w = (T - \lambda I)q(T)v = p(T)v = 0$.

5. So $Tw = \lambda w$ with $w \neq 0$.

□

Both hypotheses are needed. Complex: rotation by 90° on $\mathbb{R}^2$ has no eigenvector. Finite-dimensional: $T(p) = xp$ on $\mathcal{P}(\mathbb{C})$ has no eigenvector, since for $p \neq 0$, $\deg(xp) = \deg p + 1 \neq \deg(\lambda p)$ when $\lambda \neq 0$, and $xp \neq 0$ when $\lambda = 0$.

Theorem (Prop. 54 / LADR 5.11: eigenvectors for distinct eigenvalues are independent)
If $v_1, \dots, v_m$ are eigenvectors of T with distinct eigenvalues $\lambda_1, \dots, \lambda_m$, then $v_1, \dots, v_m$ is linearly independent.

The idea

Start with two. Could $v_2 = av_1$? Apply T to both sides: $\lambda_2 v_2 = a\lambda_1 v_1$. Multiply both sides by λ_2 instead: $\lambda_2 v_2 = a\lambda_2 v_1$. The left sides are equal, so $a(\lambda_1 - \lambda_2)v_1 = 0$, so $a = 0$, so $v_2 = 0$, which an eigenvector is not.

The trick: *applying T and multiplying by λ_k do the same thing to v_k but different things to the other eigenvectors.* Subtracting removes v_k and leaves a relation among fewer vectors, with each coefficient multiplied by a nonzero number $\lambda_i - \lambda_k$. For the general case we want the shorter relation to be among independent vectors; so pick the *first* v_k that depends on the earlier ones.

Proof. 1. Suppose the list is dependent. By (W2) there is a smallest k with $v_k \in \operatorname{span}(v_1, \dots, v_{k-1})$. Because k is smallest, $v_1, \dots, v_{k-1}$ is independent.

2. Write $v_k = a_1 v_1 + \cdots + a_{k-1} v_{k-1}$.

3. Apply T to step 2, and separately multiply step 2 by λ_k:

$$\lambda_k v_k = a_1 \lambda_1 v_1 + \cdots + a_{k-1} \lambda_{k-1} v_{k-1},$$
$$\lambda_k v_k = a_1 \lambda_k v_1 + \cdots + a_{k-1} \lambda_k v_{k-1}.$$

4. Subtracting,

$$0 = a_1(\lambda_1 - \lambda_k)v_1 + \cdots + a_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1}.$$

5. $v_1, \dots, v_{k-1}$ is independent, so each $a_i(\lambda_i - \lambda_k) = 0$. The eigenvalues are distinct, so each $a_i = 0$.

6. Then $v_k = 0$ by step 2, contradicting that eigenvectors are nonzero.

□

Prop. 55 / LADR 5.12. An operator on a finite-dimensional V has at most $\dim V$ distinct eigenvalues: eigenvectors for distinct eigenvalues are independent, and an independent list has at most $\dim V$ vectors (W1).

Example 50 and the shift: infinitely many eigenvalues. **(50)** In the space V of infinitely differentiable functions $\mathbb{R} \to \mathbb{R}$, let $T(f) = f''$. Since $(\sin kx)'' = -k^2 \sin kx$, the functions $\sin x, \sin 2x, \dots, \sin Nx$ are eigenvectors with the distinct eigenvalues $-1, -4, \dots, -N^2$, so they are independent (Prop. 54).

(Shift) In infinite dimensions an operator can have infinitely many eigenvalues. The left shift $(a_1, a_2, \dots) \mapsto (a_2, a_3, \dots)$ on $\mathbb{R}^\infty$ has every real λ as an eigenvalue, with eigenvector $(1, \lambda, \lambda^2, \dots)$: shifting gives $(\lambda, \lambda^2, \dots) = \lambda(1, \lambda, \lambda^2, \dots)$. By Prop. 54 these eigenvectors are independent, so $\mathbb{R}^\infty$ contains independent lists of every length (and in fact an independent family indexed by $\mathbb{R}$).

The question answered. Yes: with $p(z) = z^2 + 1 = (z - i)(z + i)$ and $q(z) = z + i$, the vector $w = q(T)e_1 = (i, 1)$ has $Tw = iw$ (the example after the idea box).

Shortcuts

- **A polynomial equation for T gives the candidate eigenvalues.** If $p(T) = 0$, every eigenvalue is a root of p. Reason: (iii), $0 = p(T)v = p(\lambda)v$ with $v \neq 0$. Condition: a root need not be an eigenvalue: I satisfies $P^2 = P$, and 0 is not an eigenvalue of I. Instance: $P^2 = P$ gives $\lambda \in \{0, 1\}$; $T^2 = I$ gives $\lambda = \pm 1$; $N^2 = 0$ gives $\lambda = 0$.

- n **distinct eigenvalues in dimension n: the eigenvectors form a basis.** Reason: Prop. 54 and (W1). Instance: A_3 in HW6 P5 has two distinct eigenvalues, so $(1, \lambda_+), (1, \lambda_-)$ is a basis of $\mathbb{R}^2$.
- **Inverse and shifts.** If $Tv = \lambda v$ and T is invertible, then $\lambda \neq 0$ and $T^{-1}v = \lambda^{-1}v$; and $(T+cI)v = (\lambda+c)v$. Reason: apply T^{-1} to $Tv = \lambda v$ and divide by λ ($\lambda = 0$ would give $v = T^{-1}0 = 0$). Instance: $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ has eigenvalues $1, 3$, so A^{-1} has $1, \frac{1}{3}$ and $A - 3I$ has $-2, 0$.
- **Conjugate operators have the same eigenvalues.** $S^{-1}TS$ has the eigenvalues of T, with eigenspaces $S^{-1}E(\lambda, T)$. Reason: HW6 P3. Instance: $S = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$, $T = \operatorname{diag}(1, 2)$: $S^{-1}TS = \left[\begin{smallmatrix} 1 & -1 \\ 0 & 2 \end{smallmatrix}\right]$, with eigenvalues $1, 2$.
- **Degree arguments on polynomial spaces.** Differentiation lowers degree, so $p' = \lambda p$ with $p \neq 0$ forces $\lambda = 0$. Reason: HW6 P2. Condition: on $\mathcal{P}_n$; it fails on smooth functions, where $(e^{\lambda x})' = \lambda e^{\lambda x}$.

Traps

- $p(T)v = 0$ does not make v an eigenvector. $T = \operatorname{diag}(1, 2)$, $v = (1, 1)$, $p = (z-1)(z-2)$: $p(T) = 0$, but $Tv = (1, 2)$ is not a multiple of v. Prop. 53 uses $w = q(T)v$, not v. (This was the error on Midterm 2 Q2(c).)
- Over $\mathbb{R}$ an operator may have no eigenvalue (rotation by 90°). In infinite dimensions an operator may have none even over $\mathbb{C}$: $p \mapsto xp$ on $\mathcal{P}(\mathbb{C})$.
- Independence needs *distinct* eigenvalues: v and $2v$ are eigenvectors with the same eigenvalue and are dependent.

§2.4 Computing eigenvalues (Ch. 11.2)

The question. For $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, for which λ is $A - \lambda I$ not invertible, and what are the vectors it sends to 0?

Proposition (Prop. 56: eigenvalues of an operator are eigenvalues of its matrix)
Let V be finite-dimensional with a basis $v_1, \ldots, v_n$. Then λ is an eigenvalue of T if and only if it is an eigenvalue of the matrix $\mathcal{M}(T)$, and v is an eigenvector of T for λ if and only if $\mathcal{M}(v)$ is an eigenvector of $\mathcal{M}(T)$ for λ.

Proof. 1. $\mathcal{M}(T)\mathcal{M}(v) = \mathcal{M}(Tv)$ (W5) and $\mathcal{M}(\lambda v) = \lambda\mathcal{M}(v)$.

2. Two vectors are equal exactly when their coordinate columns are equal, so $Tv = \lambda v \iff \mathcal{M}(T)\mathcal{M}(v) = \lambda\mathcal{M}(v)$.

3. $v \neq 0 \iff \mathcal{M}(v) \neq 0$, so eigenvectors correspond to eigenvectors.

□

The 2×2 characteristic polynomial. For $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$:

1. λ is an eigenvalue $\iff$ $\lambda I - A$ is not invertible $\iff$ $\det(\lambda I - A) = 0$ (LADR 5.7 and W4).
2. Apply the 2×2 formula to $tI - A$; the result is the **characteristic polynomial** $\chi_A(t)$ of A:
$$\chi_A(t) = \det(tI - A) = (t-a)(t-d) - bc = t^2 - (a+d)t + (ad - bc).$$
3. So the eigenvalues of A are the zeros of χ_A in $\mathbb{F}$. Over $\mathbb{C}$ write $\chi_A(t) = (t - \lambda_1)(t - \lambda_2)$; expanding gives $t^2 - (\lambda_1 + \lambda_2)t + \lambda_1\lambda_2$, so the two roots, *counted with multiplicity*, have sum $a + d$ and product $ad - bc$. (For $A_2 = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ the root 1 is counted twice: $1 + 1 = 2 = a + d$.)
4. Eigenvectors for λ: solve $(A - \lambda I)x = 0$. Since $\det(A - \lambda I) = 0$, the rows of $A - \lambda I$ are dependent (W4), so one is a multiple of the other, and one equation is enough. $(a - \lambda)x + by = 0$ gives $(b,\ \lambda - a)$; $cx + (d - \lambda)y = 0$ gives $(\lambda - d,\ c)$. Use whichever is nonzero. Both are zero only when $b = c = 0$ and $a = d = \lambda$, that is, $A = \lambda I$; then every nonzero vector is an eigenvector.

(Only the 2×2 formula is available for now; the characteristic polynomial in general comes later in the course.)

Example 51: rotation and Fibonacci. The rotation R and the Fibonacci matrix F have characteristic polynomials

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \chi_R = t^2 + 1; \qquad F = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \chi_F = t^2 - t - 1.$$

So R has no real eigenvalues, and over $\mathbb{C}$ its eigenvalues are $\pm i$. The eigenvalues of F are $(1 \pm \sqrt{5})/2$.

Example 52: use the structure, not a determinant. $T(A) = A^\mathsf{T}$ on $\mathbb{C}^{2,2}$ satisfies $T^2 = I$. If $TA = \lambda A$ with $A \neq 0$, apply T again: $A = T(TA) = T(\lambda A) = \lambda TA = \lambda^2 A$, so $\lambda^2 = 1$. Both values occur. $E(1, T)$ is the symmetric matrices ($A^\mathsf{T} = A$), spanned by

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix};$$

$E(-1, T)$ is the skew-symmetric matrices ($A^\mathsf{T} = -A$), spanned by $\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$.

The question answered. $\chi_A(t) = t^2 - 4t + 3 = (t-1)(t-3)$, so $A - \lambda I$ is not invertible for $\lambda = 1, 3$. For $\lambda = 3$ it sends $\operatorname{span}((1,1))$ to 0, and for $\lambda = 1$ it sends $\operatorname{span}((1,-1))$ to 0 (the figure in 2.1).

Shortcuts

- 2×2 **formula.** $\chi_A(t) = t^2 - (a+d)t + (ad - bc)$; eigenvector for λ: $(b, \lambda - a)$ or $(\lambda - d, c)$. Reason: the steps above. Condition: not both vectors zero (that happens only for $A = \lambda I$). Instance: for $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ and $\lambda = 3$, $(b, \lambda - a) = (1, 1)$.
- **Triangular 2×2: eigenvalues on the diagonal.** If $c = 0$, $\chi_A = (t-a)(t-d)$. Reason: $bc = 0$ in the formula. Instance: A_2 has the single eigenvalue 1.
- **Constant row or column sums.** If every row of A sums to s, then $A\mathbf{1} = s\mathbf{1}$, so s is an eigenvalue. If every column sums to s, then s is an eigenvalue. Reason: for columns, apply the row case to A^T and use the next item. Instance: a stochastic P has the eigenvalue $s = 1$.
- **T and T' have the same eigenvalues** (V finite-dimensional). Reason: $(T - \lambda I)' = T' - \lambda I$ (W8), and $T - \lambda I$ is not injective $\iff$ $(T - \lambda I)'$ is not surjective $\iff$ $T' - \lambda I$ is not injective (W8, W3). Instance: A and A^T have the same eigenvalues.
- **Test a proposed eigenvector with one multiplication.** Instance: $A_3(1, \lambda) = (\lambda, 1 + \lambda) = \lambda(1, \lambda)$ because $\lambda^2 = \lambda + 1$.
- **Real matrix, complex eigenvalue: conjugate.** If $Av = \lambda v$ with A real, then $A\bar{v} = \bar{\lambda}\bar{v}$, where $\bar{v}$ is the vector of conjugated coordinates. Reason: $\overline{Av} = \bar{A}\bar{v} = A\bar{v}$ because A is real and conjugation respects sums and products (W10). Instance: for A_1, $(1, -i)$ for i gives $(1, i)$ for $-i$.

Traps

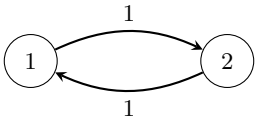
- Row reduction changes eigenvalues. $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right]$ has eigenvalues $1, 2$; it row reduces to I, whose only eigenvalue is 1.
- "Eigenvalues sum to $a + d$" counts multiplicity: A_2 has only the eigenvalue 1, and $a + d = 2$. For the real rotation there are no real eigenvalues at all, although $a + d = 0$.
- The eigenvector recipe gives 0 for $A = \lambda I$; then every nonzero vector is an eigenvector, not none.

§2.5 PageRank (Ch. 11.3)

The question. Three pages: 1 links to 2 and 3, 2 links to 3, 3 links to 1. Rank them, using the rule that a page is important when important pages link to it.

Setting up.

- Adjacency matrix: $C_{ij} = 1$ if page i links to page j, else 0. $(C^k)_{ij}$ counts directed walks of length k from i to j. (For $k = 2$: $(C^2)_{ij} = \sum_l C_{il}C_{lj}$ counts the pages l with $i \to l \to j$; induction on k does the rest.)
- Let d_j be the number of links out of page j (assume $d_j > 0$). Define $P_{ij} = 1/d_j$ if page j links to page i, else 0. Column j sums to 1, so P is stochastic. In matrix form $P = C^{\mathsf{T}}D^{-1}$ with $D = \mathrm{diag}(d_1, \ldots, d_n)$. The matrix $D^{-1}C$ is row-stochastic (each row sums to 1); books that write distributions as rows use it. In our column convention we use its transpose $P = (D^{-1}C)^{\mathsf{T}} = C^{\mathsf{T}}D^{-1}$.
- A "random surfer" on page j follows one of its links uniformly at random; if p is the distribution of the surfer's position, Pp is the distribution one step later.
- First attempt: rank by a stationary distribution $\pi = P\pi$. Coordinate i says $\pi_i = \sum_j P_{ij}\pi_j$, and $P_{ij} \neq 0$ only for pages j linking to i, each giving the share π_j/d_j: the importance of page i is collected from the pages linking to it.
- Three problems. (1) A page with no outgoing links (a **dangling page**) gives a zero column, so P is not stochastic. (2) If the web splits into parts with no links between them, there are several stationary distributions: for two pages that link only to themselves, $P = I$ and every probability vector is stationary. (3) Even with a unique stationary distribution, $P^k p_0$ need not converge: for two pages linking to each other, $P = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, the only stationary distribution is $(\frac{1}{2}, \frac{1}{2})$, but $P^k e_1$ alternates between e_1 and e_2.

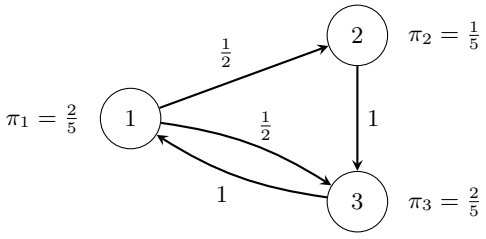


Two pages linking to each other: $P = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$. Starting at page 1, the surfer's position is $e_1, e_2, e_1, e_2, \ldots$, which does not converge, although $(\frac{1}{2}, \frac{1}{2})$ is the only stationary distribution.

The three-page web. Pages $1, 2, 3$ with links $1 \to 2$, $1 \to 3$, $2 \to 3$, $3 \to 1$. Page 1 has two links, so each carries probability $\frac{1}{2}$; pages 2 and 3 have one link each.

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 1 & 0 \end{bmatrix}.$$

The first two coordinates of $P\pi = \pi$ say $\pi_3 = \pi_1$ and $\frac{1}{2}\pi_1 = \pi_2$ (the third then holds: $\frac{1}{2}\pi_1 + \pi_2 = \pi_1 = \pi_3$). With $\pi_1 + \pi_2 + \pi_3 = 1$: $\pi = (\frac{2}{5}, \frac{1}{5}, \frac{2}{5})$.



The link graph, with the probability on each link: column j of P lists the links out of page j. In the stationary distribution, what flows into each page equals its value: page 3 receives $\frac{1}{2} \cdot \frac{2}{5} + 1 \cdot \frac{1}{5} = \frac{2}{5}$.

The PageRank equation. Fix a **damping parameter** $0 < \alpha < 1$ and a probability vector v (the **teleportation vector**). At each step the surfer follows a link with probability α, and with probability $1 - \alpha$ jumps to a page chosen according to v. A dangling column is replaced by v, so that P is stochastic. The PageRank vector r is the solution of

$$r = \alpha P r + (1 - \alpha)v, \qquad \text{equivalently} \qquad (I - \alpha P)r = (1 - \alpha)v.$$

Proposition (Prop. 57: eigenvalues of a stochastic matrix)

If P is a stochastic $n \times n$ matrix, every eigenvalue $\lambda \in \mathbb{C}$ of P satisfies $|\lambda| \leq 1$.

The idea

Measure the size of a (possibly complex) vector by $\|x\|_1 = |x_1| + \cdots + |x_n|$. The coordinates of Px are combinations of the coordinates of x with nonnegative weights that, column by column, sum to 1. So the sum of absolute values cannot grow: $\|Px\|_1 \leq \|x\|_1$. An eigenvector is scaled by $|\lambda|$, so $|\lambda| \leq 1$.

Proof. 1. By the triangle inequality and $P_{ij} \geq 0$,

$$\|Px\|_1 = \sum_i \Big|\sum_j P_{ij} x_j\Big| \leq \sum_i \sum_j P_{ij} |x_j|.$$

2. Exchange the sums; each column sums to 1, so

$$\sum_j \Big(\sum_i P_{ij}\Big) |x_j| = \sum_j |x_j| = \|x\|_1.$$

3. If $Px = \lambda x$ with $x \neq 0$: $|\lambda| \, \|x\|_1 = \|Px\|_1 \leq \|x\|_1$, and $\|x\|_1 > 0$, so $|\lambda| \leq 1$.

□

Theorem (Prop. 58: existence and uniqueness of PageRank)

For a stochastic P, a probability vector v and $0 < \alpha < 1$, there is exactly one $r \in \mathbb{R}^n$ with $r = \alpha P r + (1 - \alpha) v$, and r is a probability vector.

The idea

The equation is the linear system $(I - \alpha P) r = (1 - \alpha) v$, so everything follows if $I - \alpha P$ is invertible. For a square matrix it is enough to show injectivity (W3). And $(I - \alpha P) x = 0$ says $Px = \frac{1}{\alpha} x$: that would make $\frac{1}{\alpha} > 1$ an eigenvalue of P, which Prop. 57 forbids. Uniqueness gives existence.

To see that r is a probability vector, write the solution explicitly. Substituting the equation into itself:

$$r = (1 - \alpha) v + \alpha P r = (1 - \alpha) v + \alpha (1 - \alpha) P v + \alpha^2 P^2 r = \ldots,$$

which suggests $r = (1 - \alpha) \sum_{k \geq 0} \alpha^k P^k v$. This is a weighted average of the probability vectors $P^k v$ with weights $(1 - \alpha)\alpha^k$ that sum to 1.

Proof. 1. $I - \alpha P$ **is injective.** If $(I - \alpha P) x = 0$ with $x \neq 0$, then $Px = \frac{1}{\alpha} x$, so $\frac{1}{\alpha} > 1$ is an eigenvalue of P, contradicting Prop. 57.

2. **Unique solution.** An injective operator on $\mathbb{R}^n$ is invertible (W3). So $(I - \alpha P) r = (1 - \alpha) v$ has exactly one solution. This is the step where finite dimension turns uniqueness into existence.

3. **The series.** Let

$$s = (1 - \alpha) \sum_{k=0}^{\infty} \alpha^k P^k v.$$

Each $P^k v$ is a probability vector (W6), so each coordinate of each term lies in $[0, (1 - \alpha)\alpha^k]$, and the series converges (comparison with a geometric series, cited from analysis). The coordinates of s are ≥ 0.

4. **Sum of coordinates.**

$$\mathbf{1}^{\mathsf{T}} s = (1 - \alpha) \sum_k \alpha^k \, \mathbf{1}^{\mathsf{T}} P^k v = (1 - \alpha) \sum_k \alpha^k = 1.$$

So s is a probability vector.

5. s **solves the equation.**
$$\alpha Ps + (1-\alpha)v = (1-\alpha)\sum_{k\geq 1}\alpha^k P^k v + (1-\alpha)v = s.$$
By uniqueness $s = r$.

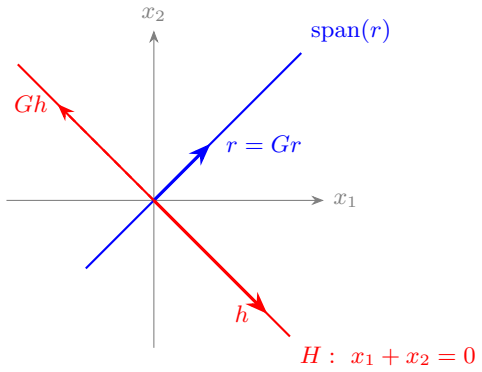
□

Computing r by iteration. Start from any probability vector r_0 and set $r_{k+1} = \alpha P r_k + (1-\alpha)v$.

1. Subtract $r = \alpha Pr + (1-\alpha)v$: $r_{k+1} - r = \alpha P(r_k - r)$.
2. By steps 1–2 of Prop. 57, $\|Px\|_1 \leq \|x\|_1$, so
$$\|r_{k+1} - r\|_1 = \alpha\|P(r_k - r)\|_1 \leq \alpha\|r_k - r\|_1, \qquad \text{so} \qquad \|r_k - r\|_1 \leq \alpha^k\|r_0 - r\|_1 \to 0.$$
The error shrinks by at least the factor α each step.
3. The map $F(x) = \alpha Px + (1-\alpha)v$ satisfies $\|F(x) - F(y)\|_1 \leq \alpha\|x - y\|_1$; a map with this property for some $\alpha < 1$ is called a **contraction**.

The Google matrix. Let $\mathbf{1} = (1, \ldots, 1)^\mathsf{T}$ and $G = \alpha P + (1-\alpha)v\mathbf{1}^\mathsf{T}$.

- For a probability vector x, $\mathbf{1}^\mathsf{T}x = 1$, so $Gx = \alpha Px + (1-\alpha)v$. The PageRank equation is $Gr = r$: r is an eigenvector of G with eigenvalue 1.
- Let $H = \{x \in \mathbb{R}^n : x_1 + \cdots + x_n = 0\}$. P keeps the sum of coordinates, since $\mathbf{1}^\mathsf{T}P = \mathbf{1}^\mathsf{T}$ (entry j of $\mathbf{1}^\mathsf{T}P$ is the sum of column j). So H is invariant under P. For $x \in H$, $\mathbf{1}^\mathsf{T}x = 0$ and $Gx = \alpha Px \in H$. So H is invariant under G, and on H the matrix G acts as αP.
- $\mathbb{R}^n = \text{span}(r) \oplus H$. H is the null space of the nonzero functional $x \mapsto x_1 + \cdots + x_n$, so $\dim H = n - 1$ (W8). A nonzero multiple cr has coordinate sum $c \neq 0$, so $\text{span}(r) \cap H = \{0\}$. The dimensions add to $1 + (n-1) = n$, so $\mathbb{R}^n = \text{span}(r) \oplus H$ (W1). On H, $\|Gx\|_1 = \alpha\|Px\|_1 \leq \alpha\|x\|_1$: G fixes r and shrinks the 1-norm of every vector of H by at least the factor α.
- History: Brin and Page used $\alpha = 0.85$ and uniform v; a non-uniform v gives personalised PageRank.



Two pages linking to each other, $\alpha = 0.85$, uniform v. G fixes $r = (\frac{1}{2}, \frac{1}{2})$, and on the invariant line H it acts as αP: $h = (1, -1) \mapsto 0.85\,(-1, 1)$. So $G^k h \to 0$ and $G^k p_0 \to r$ for every probability vector p_0, unlike $P^k p_0$.

The question answered. With $\alpha = \frac{1}{2}$ and uniform v, the PageRank vector is $r = \frac{1}{39}(14, 10, 15)$ (E1): page 3, then 1, then 2. Without damping the stationary distribution $(\frac{2}{5}, \frac{1}{5}, \frac{2}{5})$ ties pages 1 and 3.

Shortcuts

- **A stochastic matrix has the eigenvalue** 1. Reason: $P - I$ has columns summing to 0, so $\mathbf{1}^\mathsf{T}(P-I)x = 0$ and its range lies in $H \neq \mathbb{R}^n$; so $P - I$ is not surjective, hence not injective (W3). Instance: the three-page web, $\pi = (\frac{2}{5}, \frac{1}{5}, \frac{2}{5})$.
- **Uniform stationary vector.** If every row of P also sums to 1 (for example if P is symmetric), the uniform vector $u = \frac{1}{n}\mathbf{1}$ is stationary:
$$(Pu)_i = \tfrac{1}{n}\,(\text{sum of row } i) = \tfrac{1}{n}.$$

Instance: HW6 P7, $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

- **PageRank is one linear system.** Solve $(I - \alpha P)r = (1-\alpha)v$ directly for small n; iterate for large n. Reason: Prop. 58 and the iteration. Instance: two pages linking to each other with uniform v: rows of P sum to 1, so $r = (\frac{1}{2}, \frac{1}{2})$, since $\alpha Pr + (1-\alpha)v = \alpha r + (1-\alpha)r = r$.

Traps

- $|\lambda| \le 1$ for stochastic matrices is about absolute values: $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ has the eigenvalue -1 with eigenvector $(1, -1)$.
- A stationary distribution need not be unique ($P = I$), and $P^k p_0$ need not converge to it ($P = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, $p_0 = e_1$).
- Convention: in this course P has columns summing to 1 and acts on columns, P_{ij} is the probability of moving from j to i. A row-stochastic $D^{-1}C$ from another book must be transposed first.

§3 Problem set

HW6 P1. Let $P\colon V \to V$ satisfy $P^2 = P$. (a) Prove that every eigenvalue of P belongs to $\{0, 1\}$. (b) Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

Hint: (a) apply P twice to an eigenvector. (b) $v = (v - Pv) + Pv$.

HW6 P2. Let $D\colon \mathcal{P}_n(\mathbb{F}) \to \mathcal{P}_n(\mathbb{F})$ be differentiation, $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. (a) Determine all eigenvalues and eigenspaces of D. (b) Explain why the answer is different on the space of smooth functions on $\mathbb{R}$.

Hint: compare degrees in $p' = \lambda p$.

HW6 P3. Let $S\colon V \to V$ be invertible and $R = S^{-1}TS$. Prove that T and R have the same eigenvalues; more precisely $E(\lambda, R) = S^{-1}E(\lambda, T)$ for every λ.

Hint: rewrite $Rv = \lambda v$ as a statement about Sv.

HW6 P4. Let $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$. Using only the 2×2 determinant formula, prove $\det(tI - A) = t^2 - (a+d)t + (ad - bc)$. Then prove that λ is an eigenvalue of A if and only if it is a root of this polynomial.

Hint: expand det of $\left[\begin{smallmatrix} t-a & -b \\ -c & t-d \end{smallmatrix}\right]$; then use LADR 5.7 and (W4).

HW6 P5. Find the eigenvalues and eigenspaces over the indicated field, and decide whether there is a basis of eigenvectors: $A_1 = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ (over $\mathbb{R}$, then over $\mathbb{C}$), $A_2 = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ over $\mathbb{R}$, $A_3 = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ over $\mathbb{R}$.

Hint: $\chi_A(t) = t^2 - (a+d)t + (ad - bc)$; for each root solve one row of $(A - \lambda I)x = 0$; two distinct eigenvalues give a basis (Prop. 54).

HW6 P6 (optional). Let $T\colon V \to V$ and $v \neq 0$. Define $C(v, T) = \operatorname{span}(v, Tv, T^2v, \dots)$. (a) Prove that $C(v, T)$ is T-invariant. (b) If V is finite-dimensional, prove there is $m \le \dim V$ with $C(v, T) = \operatorname{span}(v, Tv, \dots, T^{m-1}v)$.

Hint: (b) take the first power $T^m v$ that depends on the earlier ones.

HW6 P7 (optional). For the stochastic matrix

$$P = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{bmatrix}$$

find a probability vector p with $Pp = p$ and explain its meaning for the three-state Markov chain.

Hint: solve $(P - I)p = 0$ and scale so the coordinates sum to 1; here the rows of P also sum to 1.

R1 (Midterm 2 review, Problem 3). For a polynomial p and an operator T with $Tv = \lambda v$, show $p(T)v = p(\lambda)v$.

Hint: $T^k v = \lambda^k v$ by induction; then add the terms of p.

R2 (Midterm 2 review, Problem 7). For any operator $T\colon V \to V$ prove that $\operatorname{null} T$ and $\operatorname{range} T$ are invariant under T, and that their annihilators are invariant under the dual map T'.

Hint: Tu is an output of T; for annihilators use $(T'\varphi)(u) = \varphi(Tu)$.

R3 (Midterm 2 review, true/false). (a) If v_1, v_2 are eigenvectors of T, then so is $v_1 + v_2$. (b) An operator on a real vector space always has an eigenvector. (c) Every vector space can be written as a direct sum of eigenspaces of some operator.

Hint: (a) $\operatorname{diag}(1, 2)$ with e_1, e_2. (b) rotation by $90°$. (c) try $T = I$.

R4 (Midterm 2, Question 2). Let V be a nonzero finite-dimensional complex vector space and $T\colon V \to V$ linear. (a) State the Fundamental Theorem of Algebra. (b) Let $v \in V$ be nonzero. Prove that there is a nonconstant polynomial $p \in \mathcal{P}(\mathbb{C})$ with $p(T)v = 0$. (c) Using the Fundamental Theorem of Algebra and (b), prove that T has an eigenvector.

Hint: (b) $v, Tv, \dots, T^n v$ are $n+1$ vectors; explain why the relation is not just $a_0 v = 0$. (c) $p(T)v = 0$ does *not* make v an eigenvector: take p of smallest degree, factor $p = (z - \lambda)q$ and use $w = q(T)v$ (Prop. 53).

R5 (Midterm 2 review, Problem 4). (a) Let $V = \operatorname{span}(1, \sin x, \cos x, \sin 2x, \cos 2x)$ over $\mathbb{R}$ and $d\colon V \to V$ differentiation. Find the eigenvalues and eigenspaces of d. (b) Let $T\colon \mathbb{C}^2 \to \mathbb{C}^2$, $T(x, y) = (y, -x)$. Find the eigenvalues and eigenvectors of T.

Hint: (a) find a polynomial p with $p(d) = 0$ on V, using $(\sin kx)'' = -k^2 \sin kx$ and $(\cos kx)'' = -k^2 \cos kx$; then use the first Shortcut of 2.3. (b) the 2×2 characteristic polynomial.

R6 (Midterm 2 extra practice (c)). Let $S, T \in \mathcal{L}(V)$ with $ST = TS$. Prove that $\operatorname{null} T$, $\operatorname{range} T$ and every eigenspace $E(\lambda, T)$ are invariant under S.

Hint: the Shortcut on commuting operators in 2.1.

E1 (PageRank by hand). For the three-page web in the text (links $1 \to 2$, $1 \to 3$, $2 \to 3$, $3 \to 1$) with $\alpha = \frac{1}{2}$ and uniform v, solve $r = \alpha P r + (1 - \alpha)v$.

Hint: write the three coordinate equations of $r = \frac{1}{2} P r + \frac{1}{6}\mathbf{1}$ and substitute.